

THE UNIVERSITY OF SYDNEY

ELEC5204 Power Systems Analysis and Protection

Lab 2: Power System Modelling and Short Circuit Calculation

—ELEC5204 lab report

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Abstract

This laboratory develops a seven-bus three-phase power system model using MATLAB/Simulink and Simscape Electrical Specialized Power Systems. The system includes synchronous generators, transformers, transmission lines and loads operating under a per-unit system. Power flow analysis is performed using the Load Flow Analyzer to determine the steady-state operating conditions of the network.

The laboratory also investigates both symmetrical and unsymmetrical fault conditions. A three-phase short circuit fault and a single line-to-ground fault are applied at Bus 3. Theoretical calculations are carried out using the impedance matrix method, Thevenin equivalent circuits and symmetrical component analysis. Sequence networks are developed to analyse the unsymmetrical fault condition.

Simulation results obtained from Simulink are compared with the theoretical calculations. The results show good agreement between analytical and simulation methods, confirming the validity of the developed power system model. The laboratory improves the understanding of power flow studies, per-unit system calculations, transformer vector groups, short-circuit analysis and sequence network modelling in practical power system applications.

The load flow results showed that the PQ bus voltages were between 0.9167 pu and 0.9259 pu. For the three-phase fault at Bus 3, the calculated fault current was 10.9194 pu. For the phase-A-to-ground fault at Bus 3, the calculated phase-A fault current was 10.9799 pu, while the phase-B and phase-C currents were approximately zero.

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1. Introduction

Modern power systems are large interconnected networks consisting of generators, transformers, transmission lines and loads. Accurate modelling and analysis of power systems are essential to ensure stable operation, reliable power delivery and effective fault protection. Power flow analysis and short-circuit analysis are two of the most important techniques used in power system studies because they provide information about system operating conditions, voltage profiles, power transfer and fault behaviour.

In this laboratory, a seven-bus three-phase power system is developed in MATLAB/Simulink using the Simscape Electrical Specialized Power Systems toolbox. The developed system includes two synchronous generators, two transformers, five transmission lines and multiple loads operating under a common per-unit system. The Load Flow Analyzer in Simulink is used to determine the steady-state operating conditions of the network, including bus voltages, voltage angles and generator power outputs.

The laboratory also investigates both symmetrical and unsymmetrical fault conditions. A three-phase short circuit fault and a single line-to-ground fault are applied at Bus 3 to study the behaviour of the power system during abnormal operating conditions. Theoretical calculations are carried out using the impedance matrix method, Thevenin equivalent circuits and symmetrical component analysis. The simulation results obtained from Simulink are compared with the theoretical calculations to verify the correctness of the developed model and analytical methods.

The main objective of this laboratory is to improve the understanding of per-unit system calculations, transformer vector group configurations, power flow analysis, sequence networks and short-circuit analysis in practical power system applications.

2. System Configuration

2.1 System Structure

The power system developed in this laboratory is a seven-bus three-phase transmission network implemented in MATLAB/Simulink using Simscape Electrical Specialized Power Systems. The network consists of two synchronous generators, two transformers, five transmission lines and five loads connected at different buses. The developed system is used for power flow studies and fault analysis under both symmetrical and unsymmetrical fault conditions.

Generator G1 is connected to Bus 1 through Transformer T1, while Generator G2 is connected to Bus 7 through Transformer T2. The transmission network operates at 230 kV and connects the buses through five transmission lines with different line lengths. Loads are connected to Bus 2, Bus 3, Bus 4, Bus 5 and Bus 6. The complete seven-bus power system model is shown in Figure 1.

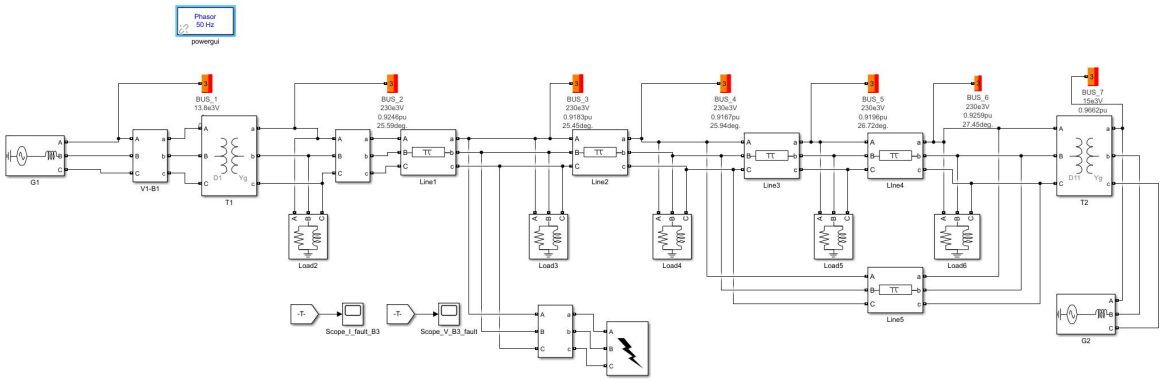


Figure 1. Complete seven-bus power system model developed in Simulink.

The model was built using Specialized Power Systems blocks. The powergui block was set to phasor simulation mode at 50 Hz. Three-Phase V-I Measurement blocks were used to measure bus voltages and branch currents in per-unit magnitude form. These measurement signals were used later to compare the Simulink fault responses with the analytical short-circuit calculations.

2.2 System Base Quantities

A common per-unit system was adopted in this laboratory because the power system contains multiple voltage levels. The selected three-phase power base is:

$$S_{base} = 100\ MVA$$

The voltage base in the G1 voltage zone is:

$$V_{base,G1} = 13.8\ kV$$

Therefore, the base impedance in the G1 voltage zone is calculated as:

$$Z_{base,G1} = \frac{V_{base,G1}^2}{S_{base}} = \frac{(13.8\ kV)^2}{100\ MVA} = 1.9044\ \Omega$$

The transmission network operates at 230 kV. Therefore, the voltage base in the transmission voltage zone is:

$$V_{base,230kV} = 230\ kV$$

The corresponding base impedance in the 230 kV transmission zone is:

$$Z_{base,230kV} = \frac{V_{base,230kV}^2}{S_{base}} = \frac{(230\ kV)^2}{100\ MVA} = 529.0000\ \Omega$$

The current base in the 230 kV transmission zone is:

$$I_{base,230kV} = \frac{S_{base}}{\sqrt{3}V_{base,230kV}} = \frac{100\ MVA}{\sqrt{3} \times 230\ kV} = 251.0219\ A$$

For the G2 voltage zone, the voltage base is:

$$V_{base,G2} = 15.0\ kV$$

These base quantities were used to convert the generator, transformer, transmission line and load parameters into the common per-unit system before performing the load flow and short-circuit calculations.

3. Task 1 – Power System Modelling

3.1 Generator Parameters

Two generators are used in the seven-bus system. Generator G1 is connected to Bus 1 and Generator G2 is connected to Bus 7. Both generators are modelled using Three-Phase Source blocks in Simulink. The generator internal resistance was neglected because only reactance data were provided in the lab specification.

Generator G1 is rated at 100 MVA and 13.8 kV. Since the rating of G1 is the same as the selected system base, its per-unit reactance values do not need to be converted to a new MVA base.

$$S_{G1} = 100\ MVA$$
$$V_{G1} = 13.8\ kV$$
$$x''_{G1} = 0.12\ pu$$
$$x_{2,G1} = 0.14\ pu$$
$$x_{0,G1} = 0.05\ pu$$

The base impedance in the G1 voltage zone is:

$$Z_{base,G1} = 1.9044\ \Omega$$

The sub-transient reactance of G1 in ohms is:

$$X''_{G1} = x''_{G1} Z_{base,G1} = 0.12 \times 1.9044 = 0.2285\ \Omega$$

The corresponding source inductance used in the Simulink Three-Phase Source block is:

$$L_{G1} = \frac{X''_{G1}}{2\pi f} = \frac{0.2285}{2\pi \times 50} = 0.727 \times 10^{-3}\ H$$

Therefore, the source resistance of G1 was set to 0 Ω and the source inductance was set to 0.727e-3 H. The neutral of G1 is solidly grounded, and the source configuration was selected as Yg.

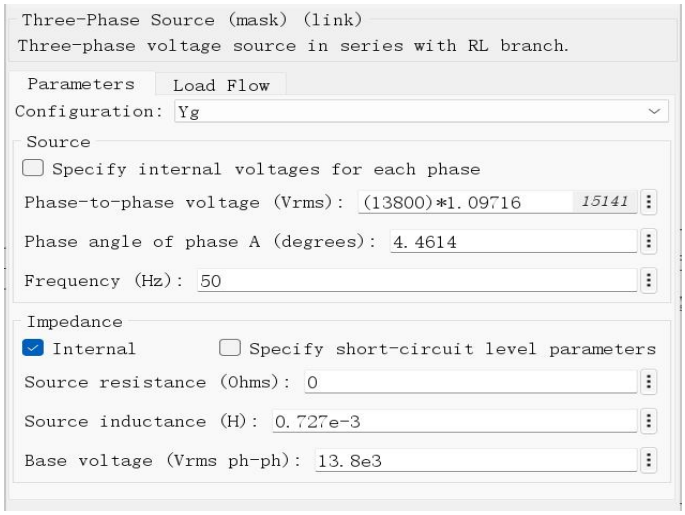


Figure 2. Parameter settings of Generator G1.

Generator G2 is rated at 200 MVA and 15.0 kV. Since the selected system base is 100 MVA, the generator reactances must be converted from the generator base to the system base.

$$S_{G2,rated} = 200\ MVA$$

$$V_{G2} = 15.0\ kV$$

$$x''_{G2,old} = 0.12\ pu$$

$$x_{2,G2,old} = 0.14\ pu$$

$$x_{0,G2,old} = 0.05\ pu$$

The per-unit conversion formula is:

$$x_{new} = x_{old}\frac{S_{base,new}}{S_{base,old}}\left(\frac{V_{base,old}}{V_{base,new}}\right)^2$$

Since the voltage base is the same as the generator rated voltage, the voltage ratio is equal to 1. Therefore:

$$x''_{G2,new} = 0.12 \times \frac{100}{200} = 0.0600\ pu$$

$$x_{2,G2,new} = 0.14 \times \frac{100}{200} = 0.0700\ pu$$

$$x_{0,G2,new} = 0.05 \times \frac{100}{200} = 0.0250\ pu$$

The base impedance in the G2 voltage zone is:

$$Z_{base,G2} = \frac{V_{base,G2}^2}{S_{base}} = \frac{(15.0\ kV)^2}{100\ MVA} = 2.2500\ \Omega$$

The sub-transient reactance of G2 in ohms is:

$$X''_{G2} = x''_{G2,new} Z_{base,G2} = 0.0600 \times 2.2500 = 0.1350\ \Omega$$

The corresponding source inductance used in the Simulink Three-Phase Source block is:

$$L_{G2} = \frac{X''_{G2}}{2\pi f} = \frac{0.1350}{2\pi \times 50} = 0.4297 \times 10^{-3}\ H$$

Therefore, the source resistance of G2 was set to 0 Ω and the source inductance was set to 0.4297e-3 H. The neutral of G2 is solidly grounded, and the source configuration was selected as Yg.

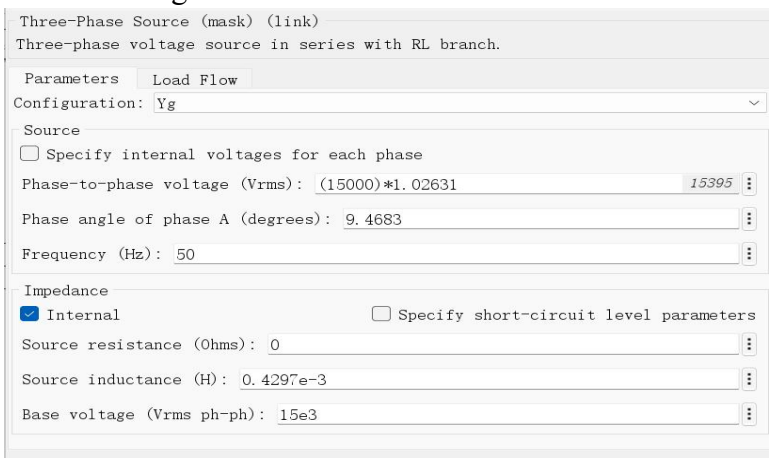


Figure 3. Parameter settings of Generator G2.

In the load flow setting, Generator G2 was configured as a PV generator. The active power generation was set to 180 MW, and the reactive power limits were set to -87 Mvar and 87 Mvar.

$$P_{G2} = 180\ MW$$

$$Q_{G2,min} = -87\ Mvar$$

$$Q_{G2,max} = 87\ Mvar$$

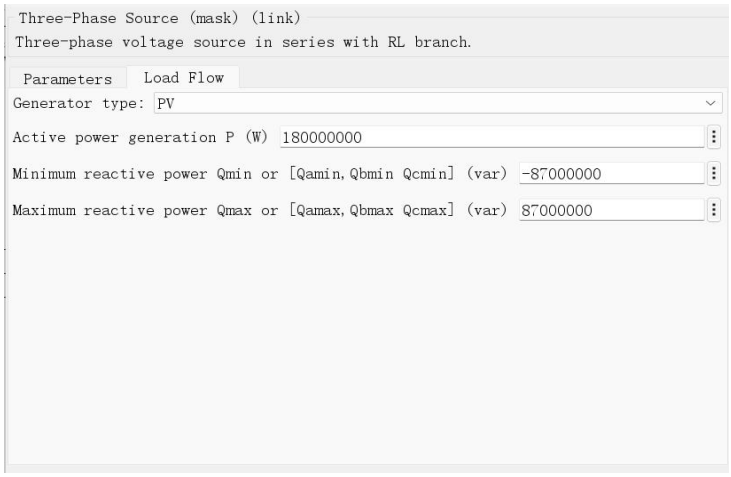


Figure 4. Load flow settings of Generator G2 as a PV bus.

3.2 Transformer Parameters

Two three-phase two-winding transformers are used in the system. Transformer T1 connects Generator G1 to the 230 kV transmission network, and Transformer T2 connects Generator G2 to the 230 kV transmission network. Both transformers were configured using the Three-Phase Transformer (Two Windings) block. The winding connected to the generator side was set as Delta (D1), and the winding connected to the transmission network side was set as grounded Wye (Yg). The core type was selected as Three single-phase transformers.

The single-phase equivalent circuit of the transformer model is shown in Figure 5. In this model, R_1 and L_1 represent the winding resistance and leakage inductance of winding 1, while R_2 and L_2 represent the winding resistance and leakage inductance of winding 2. R_m and L_m represent the magnetising branch. Since only transformer reactance was provided in the lab specification, the winding resistances were set to zero. The magnetisation resistance and magnetisation inductance were set to large values so that the transformer behaves close to an ideal transformer while still retaining the required model structure.

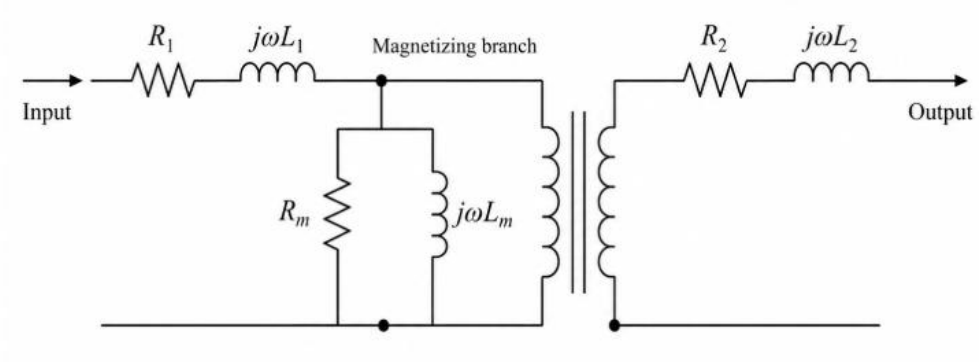


Figure 5. Single-phase equivalent circuit of the transformer model.

Transformer T1 is rated at 100 MVA with a voltage ratio of 13.8 kV / 230 kV. The transformer leakage reactance is given as 0.1 pu on the transformer rating. Since the rating of T1 is the same as the selected system base, the transformer reactance does not need to be converted to a new MVA base.

$$S_{T1,rated} = 100\ MVA$$

$$V_{T1} = 13.8\ kV/230\ kV$$

$$x_{T1,old} = 0.1\ pu$$

$$x_{T1,new} = x_{T1,old}\frac{S_{base,new}}{S_{base,old}}$$

$$x_{T1,new} = 0.1 \times \frac{100}{100}$$

$$x_{T1,new} = 0.1\ pu$$

In the Simulink transformer block, the total leakage reactance was split equally between the two windings. Therefore, the winding leakage inductance values were set as:

$$L_{1,T1} = 0.05\ pu$$
$$L_{2,T1} = 0.05\ pu$$
$$R_{1,T1} = 0\ pu$$
$$R_{2,T1} = 0\ pu$$

The magnetisation resistance and magnetisation inductance were set as:

$$R_{m,T1} = 500\ pu$$
$$L_{m,T1} = 500\ pu$$

The connection and parameter settings of T1 are shown in Figure 6 and Figure 7.

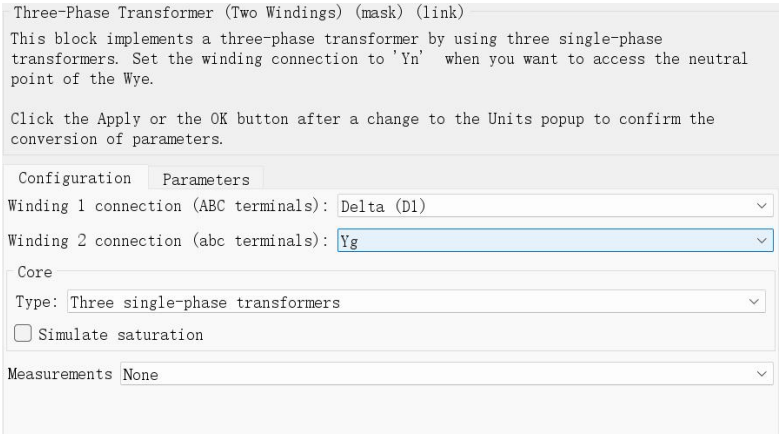


Figure 6. Winding connection settings of Transformer T1.

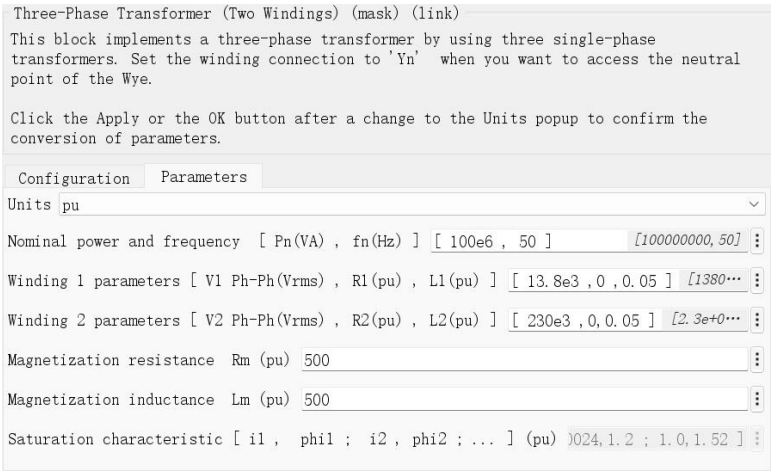


Figure 7. Parameter settings of Transformer T1.

Transformer T2 is rated at 200 MVA with a voltage ratio of 15.0 kV / 230 kV. The transformer leakage reactance is given as 0.1 pu on the transformer rating. Since the selected system base is 100 MVA, the reactance of T2 must be converted to the system base for the analytical short-circuit calculations.

$$S_{T2,rated} = 200\ MVA$$
$$V_{T2} = 15.0\ kV/230\ kV$$
$$x_{T2,old} = 0.1\ pu$$

$$x_{T2,new} = x_{T2,old}\frac{S_{base,new}}{S_{base,old}}$$
$$x_{T2,new} = 0.1 \times \frac{100}{200}$$
$$x_{T2,new} = 0.05\ pu$$

Therefore, the equivalent leakage reactance of T2 is 0.05 pu on the 100 MVA system base. In the Simulink transformer block, the transformer rated power was entered as 200 MVA. The leakage reactance was split equally between the two windings on the transformer block base:

$$L_{1,T2} = 0.05\ pu$$
$$L_{2,T2} = 0.05\ pu$$
$$R_{1,T2} = 0\ pu$$
$$R_{2,T2} = 0\ pu$$

The magnetisation branch values were set as:

$$R_{m,T2} = 500\ pu$$

$$L_{m,T2} = 500\ pu$$

The connection and parameter settings of T2 are shown in Figure 8 and Figure 9.

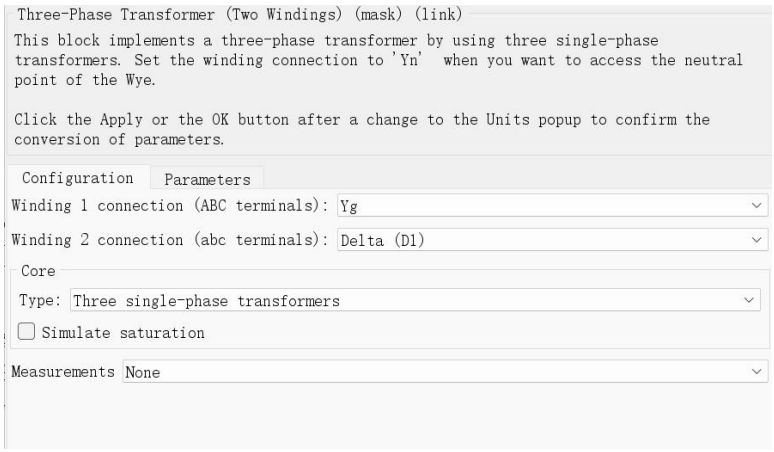


Figure 8. Winding connection settings of Transformer T2.

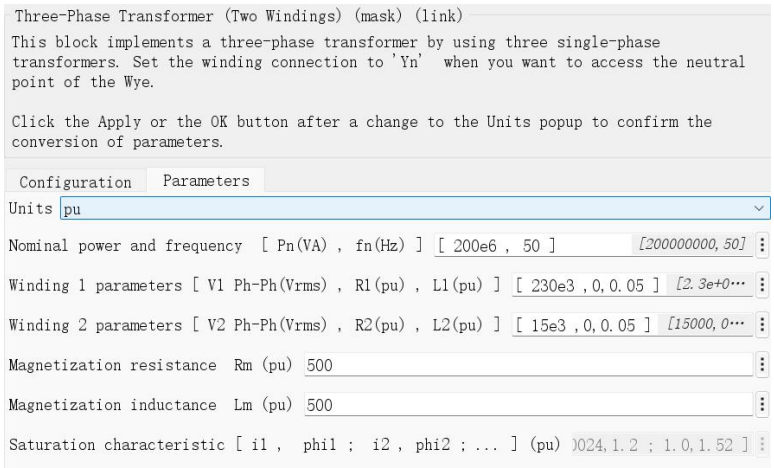


Figure 9. Parameter settings of Transformer T2.

3.3 Transmission Line Parameters

Five three-phase transmission lines are used in the seven-bus power system. All transmission lines operate at the 230 kV voltage level. The line parameters are given in positive-sequence and zero-sequence form as:

$$z_1 = 0.08 + j0.5\ \Omega/km$$

$$z_0 = 0.2 + j1.5\ \Omega/km$$

$$y_1 = j3.3 \times 10^{-6}\ S/km$$

The frequency of the system is 50 Hz. Therefore, the positive-sequence and zero-sequence inductances per kilometre are calculated from the reactance values:

$$L_1 = \frac{X_1}{2\pi f} = \frac{0.5}{2\pi \times 50} = 1.592 \times 10^{-3}\ H/km$$

$$L_0 = \frac{X_0}{2\pi f} = \frac{1.5}{2\pi \times 50} = 4.775 \times 10^{-3}\ H/km$$

The positive-sequence capacitance per kilometre is obtained from the shunt admittance:

$$C_1 = \frac{B_1}{2\pi f} = \frac{3.3 \times 10^{-6}}{2\pi \times 50} = 10.5 \times 10^{-9}\ F/km$$

Since the zero-sequence capacitance is not given in the laboratory specification, it is set equal to the positive-sequence capacitance:

$$C_0 = C_1 = 10.5 \times 10^{-9}\ F/km$$

Therefore, the same per-kilometre line parameters were entered into each Three-Phase PI Section Line block:

$$r_1 = 0.08\ \Omega/km$$

$$r_0 = 0.2\ \Omega/km$$

$$L_1 = 1.592 \times 10^{-3}\ H/km$$

$$L_0 = 4.775 \times 10^{-3}\ H/km$$

$$C_1 = C_0 = 10.5 \times 10^{-9}\ F/km$$

The only parameter that changes between different line blocks is the line length. The line lengths are 15 km, 25 km, 40 km, 15 km and 50 km respectively. The parameter settings of Line 1 are shown in Figure 10.

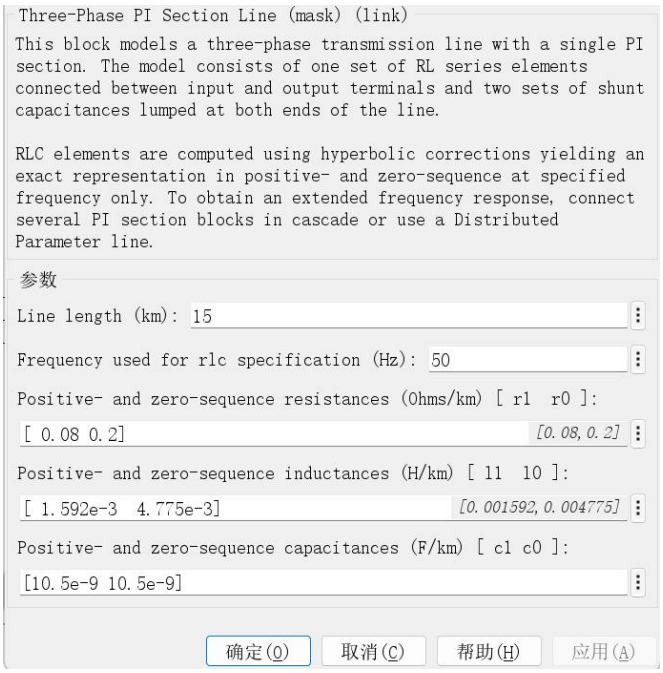


Figure 10. Parameter settings of transmission line Line 1.

The total line impedance is calculated by multiplying the per-kilometre impedance by the line length:

$$Z_1 = z_1 l$$

$$Z_0 = z_0 l$$

The per-unit impedance is obtained using the 230 kV base impedance:

$$Z_{base,230kV} = 529.0000\ \Omega$$

$$Z_{pu} = \frac{Z_\Omega}{Z_{base,230kV}}$$

The total positive-sequence shunt admittance is calculated as:

$$Y_1 = j3.3 \times 10^{-6} l$$

The corresponding per-unit shunt admittance is:

$$Y_{1,pu} = Y_1 Z_{base,230kV}$$

The calculated transmission line parameters are summarised in Table 2.

Table 2. Calculated transmission line impedances and shunt admittances.

Line	**Length km**	**$Z_1\Omega$**	**$Z_{1,pu}$**	**$Z_0\Omega$**	**$Z_{0,pu}$**	**$Y_{1,pu}$**
Line 1	15	1.2 + j7.5	0.00227 + j0.01418	3.0 + j22.5	0.00567 + j0.04253	j0.02619
Line 2	25	2.0 + j12.5	0.00378 + j0.02363	5.0 + j37.5	0.00945 + j0.07089	j0.04364
Line 3	40	3.2 + j20.0	0.00605 + j0.03781	8.0 + j60.0	0.01512 + j0.11342	j0.06983
Line 4	15	1.2 + j7.5	0.00227 + j0.01418	3.0 + j22.5	0.00567 + j0.04253	j0.02619
Line 5	50	4.0 + j25.0	0.00756 + j0.04726	10.0 + j75.0	0.01890 + j0.14178	j0.08729

The same per-kilometre resistance, inductance and capacitance values were used for Line 2 to Line 5. The only changed parameter was the line length. The parameter settings of Line 2 to Line 5 are shown in Figure 11.

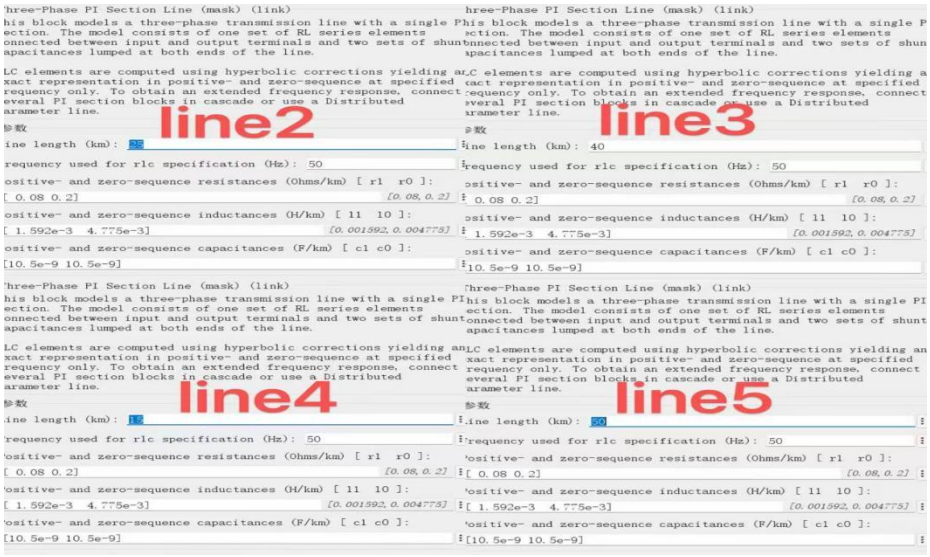


Figure 11. Parameter settings of transmission lines Line 2 to Line 5.

3.4 Load Parameters

Five balanced three-phase loads are connected to Bus 2, Bus 3, Bus 4, Bus 5 and Bus 6. Each load was modelled using a Three-Phase Parallel RLC Load block with grounded wye connection. According to the lab system data, each load consumes 50 MW active power and 30 Mvar inductive reactive power at the 230 kV transmission voltage level.

The complex power of each load is:

$$S_{load} = P_{load} + jQ_{load}$$

$$S_{load} = 50\ MW + j30\ Mvar$$

Using the selected three-phase system base:

$$S_{base} = 100\ MVA$$

the per-unit load power is:

$$S_{load,pu} = \frac{S_{load}}{S_{base}}$$

$$S_{load,pu} = \frac{50 + j30}{100}$$

$$S_{load,pu} = 0.5 + j0.3\ pu$$

Therefore, each load was set with an active power of 50e6 W and an inductive reactive power of 30e6 var. The capacitive reactive power was set to zero because the load is inductive. The nominal phase-to-phase voltage was set to 230 kV and the frequency was set to 50 Hz.

The parameter settings of one representative load are shown in Figure 12. The same settings were applied to all loads connected to Bus 2 to Bus 6.

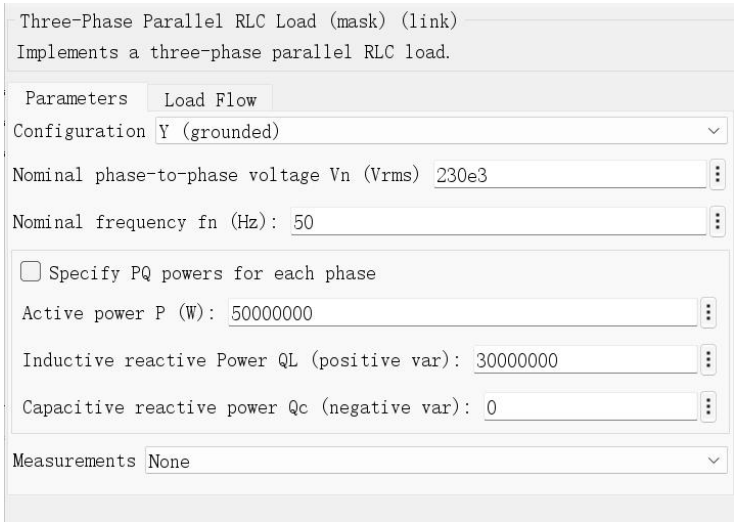


Figure 12. Parameter settings of the three-phase load connected to Bus 2.

The load settings used in the model are summarised in Table 3.

Table 3. Load settings used for Bus 2 to Bus 6.

Load	Connected Bus	(P) MW	$QMvar$	S_{pu}	Voltage kV	Connection
Load2	BUS2	50	30	0.5 + j0.3	230	Y grounded
Load3	BUS3	50	30	0.5 + j0.3	230	Y grounded
Load4	BUS4	50	30	0.5 + j0.3	230	Y grounded
Load5	BUS5	50	30	0.5 + j0.3	230	Y grounded
Load6	BUS6	50	30	0.5 + j0.3	230	Y grounded

3.5 Measurement Block Configuration

Three-Phase V-I Measurement blocks were used to measure bus voltages and branch currents in the Simulink model. These measurement blocks are important because the analytical short-circuit calculation results need to be compared with the simulation outputs.

For voltage measurement, the measurement type was selected as phase-to-ground voltage. This is because the fault analysis in Task 3 and Task 4 requires the phase voltage at

the faulted bus and other buses. The output signal was selected as magnitude so that the simulation results can be directly compared with the calculated per-unit voltage magnitudes.

For the transmission network, the base voltage and base power used in the measurement blocks were:

$$V_{base,230kV} = 230\ kV$$

$$S_{base} = 100\ MVA$$

Therefore, the current base at the 230 kV level is:

$$I_{base,230kV} = \frac{S_{base}}{\sqrt{3}V_{base,230kV}} = \frac{100\ MVA}{\sqrt{3} \times 230\ kV} = 251.0219\ A$$

The measurement block connected near Bus 3 was used to observe the voltage response and fault current during the three-phase fault and the phase-A-to-ground fault. The measured voltage and current signals were connected to Scope blocks to obtain the simulation waveforms.

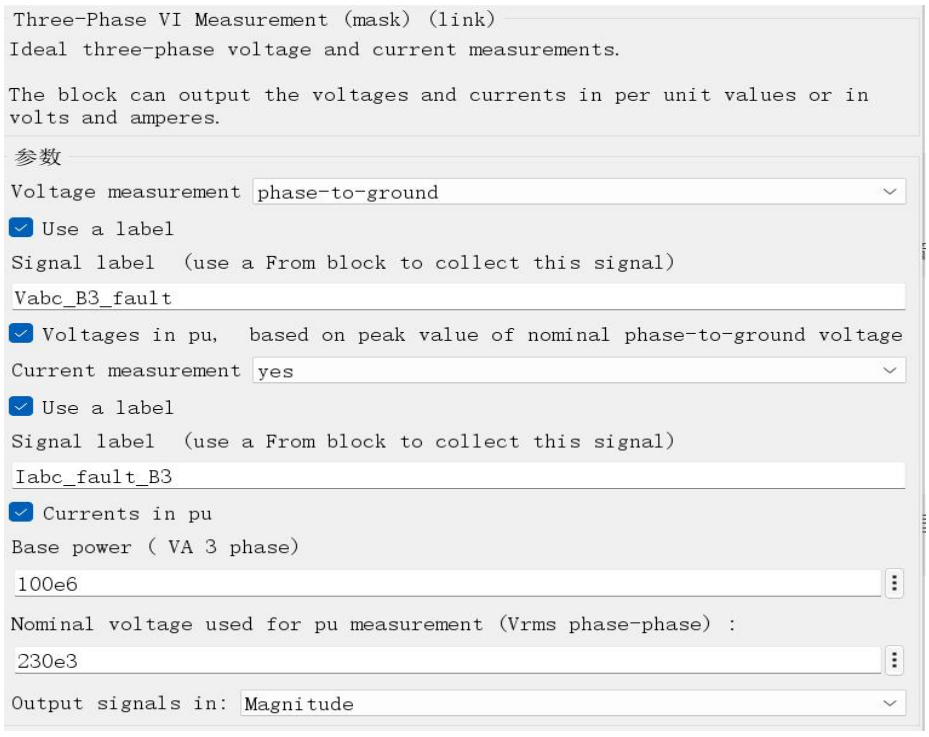


Figure 13. Representative settings of the Three-Phase V-I Measurement block.

4. Task 2 – Load Flow Analysis

4.1 Bus Type Settings

There are seven buses in the studied power system. Bus 1 is selected as the swing bus because it is connected to Generator G1, which provides the reference voltage angle for the system. Bus 7 is selected as the PV bus because Generator G2 controls the voltage magnitude at this bus while generating a specified active power. Bus 2 to Bus 6 are selected as PQ buses because they are load buses with specified active and reactive power demand.

The bus type settings are summarised in Table 4.

Table 4. Bus type settings used in the load flow study.

Bus	Bus type	Main reason
BUS1	Swing	Reference bus connected to G1, with V = 1.0 pu and angle = 0 deg
BUS2	PQ	Load bus with specified P and Q demand
BUS3	PQ	Load bus with specified P and Q demand
BUS4	PQ	Load bus with specified P and Q demand
BUS5	PQ	Load bus with specified P and Q demand
BUS6	PQ	Load bus with specified P and Q demand
BUS7	PV	Generator bus connected to G2, with specified P and voltage magnitude

For Generator G1, the generator type was selected as swing bus. For Generator G2, the generator type was selected as PV bus. The active power generation of G2 was set to 180 MW, and its reactive power limits were set between -87 Mvar and 87 Mvar.

$$P_{G2} = 180 \text{ MW}$$

$$Q_{G2,min} = -87 \text{ Mvar}$$

$$Q_{G2,max} = 87 \text{ Mvar}$$

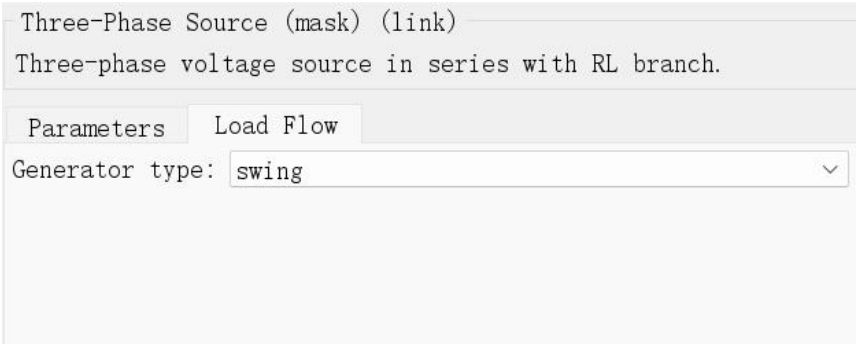


Figure 14. Load flow setting of Generator G1 as the swing bus.

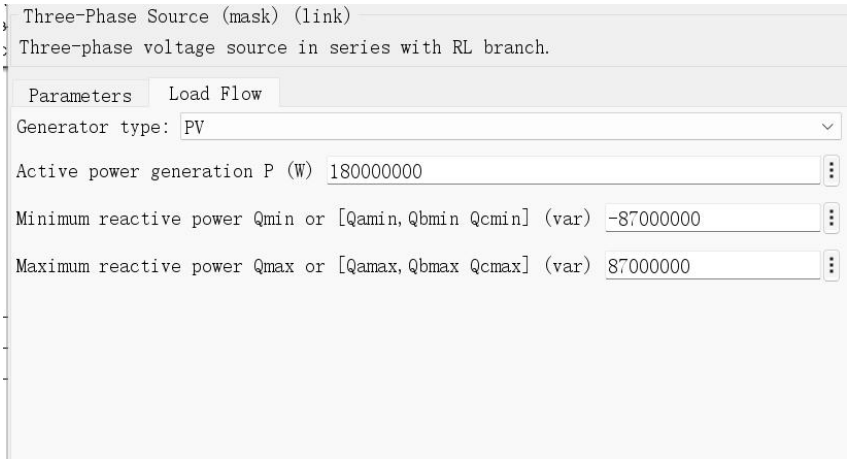


Figure 15. Load flow setting of Generator G2 as the PV bus.

4.2 Load Flow Input Data Check

Before running the load flow calculation, the input data table in the Load Flow Analyzer was checked carefully. This step is necessary because incorrect bus type, bus ID, voltage base, voltage reference, active power, reactive power, or generator reactive power limits may lead to non-convergence or incorrect load flow results. The lab instruction also requires the input data table to be checked before starting the calculation.

The main checked quantities are expressed using the following notation:

$V_{base,i}$: base voltage of bus i

$V_{ref,i}$: specified reference voltage magnitude of bus i

$\theta_{ref,i}$: specified reference voltage angle of bus i

P_i: specified active power injection or demand at bus i

Q_i: specified reactive power injection or demand at bus i

$Q_{min,G2}$: minimum reactive power limit of Generator G2

$Q_{max,G2}$: maximum reactive power limit of Generator G2

For the swing bus, BUS1, the specified voltage reference was checked as:

$$V_{base,1} = 13.8\ kV$$

$$V_{ref,1} = 1.0000\ pu$$

$$\theta_{ref,1} = 0°$$

For the PV bus, BUS7, the specified voltage magnitude and active power generation were checked as:

$$V_{base,7} = 15.0\ kV$$

$$V_{ref,7} = 1.0000\ pu$$

$$P_{G2} = 180\ MW$$

The reactive power limits of Generator G2 were checked as:

$$Q_{min,G2} = -87\ Mvar$$

$$Q_{max,G2} = 87\ Mvar$$

For the PQ load buses, BUS2 to BUS6, the voltage base and load powers were checked as:

$$V_{base,i} = 230\ kV, \quad i = 2,3,4,5,6$$

$$P_{load,i} = 50\ MW, \quad i = 2,3,4,5,6$$

$$Q_{load,i} = 30\ Mvar, \quad i = 2,3,4,5,6$$

Using the 100 MVA system base, the per-unit load powers were checked as:

$$P_{load,i,pu} = \frac{P_{load,i}}{S_{base}} = \frac{50\ MW}{100\ MVA} = 0.5\ pu$$

$$Q_{load,i,pu} = \frac{Q_{load,i}}{S_{base}} = \frac{30\ Mvar}{100\ MVA} = 0.3\ pu$$

Therefore, the complex power of each PQ load bus was checked as:

$$S_{load,i,pu} = 0.5 + j0.3\ pu, \quad i = 2,3,4,5,6$$

The input data table in the Load Flow Analyzer was checked using the following items:

1. The bus type column was checked to confirm that BUS1 was set as the swing bus, BUS7 was set as the PV bus, and BUS2 to BUS6 were set as PQ buses.

2. The bus ID column was checked to confirm that each generator and load was assigned to the correct bus.

3. The base voltage column was checked to confirm that the correct voltage bases were used:

$$V_{base,1} = 13.8\ kV$$

$$V_{base,i} = 230\ kV, \quad i = 2,3,4,5,6$$

$$V_{base,7} = 15.0\ kV$$

4. The voltage reference and angle columns were checked to confirm that the reference settings were correct

$$V_{ref,1} = 1.0000\ pu$$

$$\theta_{ref,1} = 0°$$

$$V_{ref,7} = 1.0000\ pu$$

5. The active and reactive power columns were checked to confirm that each load bus was set to 50 MW and 30 Mvar, and that Generator G2 was set to generate 180 MW.

6. The reactive power limit columns were checked to confirm that Generator G2 had the correct reactive power limits:

$$Q_{min,G2} = -87\ Mvar$$

$$Q_{max,G2} = 87\ Mvar$$

After checking the input data, the load flow calculation was performed in the Load Flow Analyzer. The converged results are reported in the next section.

	Block name	Block type	Bus type	Bus ID	Vbase (kV)	Vref (pu)	Vangle (deg)	P (MW)	Q (Mvar)	Qmin (Mvar)	Qmax (Mvar)
1	Load2	RLC load	PQ	BUS_2	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf
2	G1	Vsrc	swing	BUS_1	13.8000	1.0000	0	0.0100	0	-Inf	Inf
3	G2	Vsrc	PV	BUS_7	15.0000	1.0000	0	180.0000	0	-87.0000	87.0000
4	Load3	RLC load	PQ	BUS_3	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf
5	Load4	RLC load	PQ	BUS_4	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf
6	Load6	RLC load	PQ	BUS_6	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf
7	Load5	RLC load	PQ	BUS_5	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf

Figure 17. Load flow input data table checked before running the calculation.

4.3 Load Flow Results under D1-Yg Transformer Connection

After the load flow input data were checked, the load flow calculation was performed in the Load Flow Analyzer. The calculation converged successfully. The solved bus voltage magnitudes and phase angles were used as the pre-fault operating point for the later short-circuit calculations in Task 3 and Task 4.

In this section, the solved load flow voltage magnitude and voltage angle of bus i are denoted as:

$$|V_{LF,i}|$$

$$\theta_{LF,i}$$

The active and reactive power outputs of generator k are denoted as:

$$P_{LF,Gk}$$

$$Q_{LF,Gk}$$

The converged load flow result from Simulink is shown in Figure 18.

V_LF (pu)	Vangle_LF (deg)	P_LF (MW)	Q_LF (MVA)
0.9246	25.5915	50.0000	30.0000
1.0000	0	71.1627	78.2398
0.9662	3.2158	180.0000	87.0015
0.9183	25.4490	50.0000	30.0000
0.9167	25.9447	50.0000	30.0000
0.9259	27.4477	50.0000	30.0000
0.9196	26.7216	50.0000	30.0000

Figure 18. Converged load flow results under the D1-Yg transformer connection.

The solved bus voltage magnitudes and phase angles are summarised in Table 5.

Table 5. Load flow bus voltage results under the D1-Yg transformer connection.

Bus	$\|V_{LF,i}\|pu$	$\theta_{LF,i}deg$
BUS1	1.0000	0.0000
BUS2	0.9246	25.5915
BUS3	0.9183	25.4490
BUS4	0.9167	25.9447
BUS5	0.9196	26.7216
BUS6	0.9259	27.4477
BUS7	0.9662	3.2158

The generator active and reactive power outputs are summarised in Table 6.

Table 6. Generator active and reactive power outputs from the load flow result.

Generator	Bus	Bus type	$P_{LF,Gk}MW$	$Q_{LF,Gk}Mvar$
G1	BUS1	Swing	71.1627	78.2398
G2	BUS7	PV	180.0000	87.0015

The result shows that BUS1 remains at the reference voltage of 1.0000 pu and 0 degree because it is the swing bus. The load buses BUS2 to BUS6 have voltage magnitudes lower than 1.0 pu due to the voltage drops across the transformers and transmission lines. BUS3 has a pre-fault voltage magnitude of 0.9183 pu and an angle of 25.4490 degrees. This value is important because it is used as the Thevenin open-circuit voltage for the three-phase fault calculation at BUS3.

$$V_{th} = V_{pre-fault,3}$$

$$V_{th} = 0.9183\angle 25.4490^{\circ}\ pu$$

Generator G1 supplies the remaining active and reactive power required by the system because it is the swing generator. Generator G2 generates the specified active power of 180 MW. Its reactive power output is approximately equal to the upper reactive power limit, which indicates that G2 is operating close to its maximum reactive power capability.

$$Q_{LF,G2} \approx Q_{max,G2}$$
$$Q_{LF,G2} \approx 87.0\ Mvar$$

4.4 Effect of Transformer Winding Connection

In this section, the effect of transformer winding connection on the load flow results is investigated. The original model uses D1-Yg transformer connections, where the generator-side winding is connected in Delta and the transmission-network-side winding is connected in grounded Wye. To compare the effect of transformer phase shift, the low-voltage sides of T1 and T2 were changed to D11 and then to Yg. The load flow calculation was repeated for each case.

The three transformer connection cases are:

Case 1: D1-Yg connection

Case 2: D11-Yg connection

Case 3: Yg-Yg connection

For the D1-Yg case, the transformer connection was kept as the original setting, and the converged load flow results were reported in Section 4.3.

For the D11-Yg case, the low-voltage generator-side windings of T1 and T2 were changed from D1 to D11. The resulting load flow results are shown in Figure 19.

Bus ID	Vbase (kV)	Vref (pu)	Vangle (deg)	P (MW)	Q (Mvar)	Qmin (Mvar)	Qmax (Mvar)	V_LF (pu)	Vangle_LF (deg)	P_LF (MW)	Q_LF (MVA)
BUS_2	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9246	-34.4085	50.0000	30.0000
BUS_1	13.8000	1.0000	0	0.0100	0	-Inf	Inf	1.0000	0	71.1627	78.2398
BUS_7	15.0000	1.0000	0	180.0000	0	-87.0000	87.0000	0.9662	3.2158	180.0000	87.0015
BUS_3	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9183	-34.5510	50.0000	30.0000
BUS_4	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9167	-34.0553	50.0000	30.0000
BUS_6	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9259	-32.5523	50.0000	30.0000
BUS_5	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9196	-33.2784	50.0000	30.0000

Figure 19. Load flow results under the D11-Yg transformer connection.

For the Yg-Yg case, both windings of T1 and T2 were changed to grounded Wye connections. The resulting load flow results are shown in Figure 20.

Bus ID	Vbase (kV)	Vref (pu)	Vangle (deg)	P (MW)	Q (Mvar)	Qmin (Mvar)	Qmax (Mvar)	V_LF (pu)	Vangle_LF (deg)	P_LF (MW)	Q_LF (MVA)
BUS_2	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9246	-4.4085	50.0000	30.0000
BUS_1	13.8000	1.0000	0	0.0100	0	-Inf	Inf	1.0000	0	71.1627	78.2398
BUS_7	15.0000	1.0000	0	180.0000	0	-87.0000	87.0000	0.9662	3.2158	180.0000	87.0015
BUS_3	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9183	-4.5510	50.0000	30.0000
BUS_4	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9167	-4.0553	50.0000	30.0000
BUS_6	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9259	-2.5523	50.0000	30.0000
BUS_5	230.0000	1.0000	0	50.0000	30.0000	-Inf	Inf	0.9196	-3.2784	50.0000	30.0000

Figure 20. Load flow results under the Yg-Yg transformer connection.

The solved voltage magnitudes and phase angles under the three transformer connection cases are compared in Table 7.

Table 7. Comparison of bus voltage phase angles under different transformer winding connections.

Bus	$\mathbf{D1-Yg(\theta_{LF,i})deg}$	$\mathbf{D11-Yg(\theta_{LF,i})deg}$	$\mathbf{Yg-Yg(\theta_{LF,i})deg}$
BUS1	0.0000	0.0000	0.0000
BUS2	25.5915	-34.4085	-4.4085
BUS3	25.4490	-34.5510	-4.5510
BUS4	25.9447	-34.0553	-4.0553
BUS5	26.7216	-33.2784	-3.2784
BUS6	27.4477	-32.5523	-2.5523
BUS7	3.2158	3.2158	3.2158

From Table 7, it can be observed that the transformer winding connection has little influence on the voltage magnitudes. For BUS2 to BUS6, the voltage magnitudes remain the same in the D1-Yg, D11-Yg and Yg-Yg cases. For example, the voltage magnitude at BUS3 remains 0.9183 pu in all three cases. Therefore, the steady-state voltage magnitude profile is not significantly affected by changing the transformer vector group.

However, the voltage phase angles change clearly when the transformer connection is changed. Compared with the Yg-Yg case, the D1-Yg case shifts the voltage angles of BUS2 to BUS6 by approximately +30 degrees. For example, the voltage angle at BUS3 changes from -4.5510 degrees in the Yg-Yg case to 25.4490 degrees in the D1-Yg case:

$$\theta_{LF,3,D1} - \theta_{LF,3,Yg} = 25.4490° - (-4.5510°) = 30.0000°$$

Similarly, compared with the Yg-Yg case, the D11-Yg case shifts the voltage angles of BUS2 to BUS6 by approximately -30 degrees. For BUS3:

$$\theta_{LF,3,D11} - \theta_{LF,3,Yg} = -34.5510° - (-4.5510°) = -30.0000°$$

Therefore, the results show that the transformer vector group mainly affects the voltage phase-angle relationship, while the voltage magnitudes and generator power outputs remain almost unchanged. This is consistent with transformer vector group theory, where D1 and

D11 introduce opposite phase displacement directions, while the Yg-Yg connection does not introduce the same Delta-Wye phase shift.

According to the North American standard used in the class, the high-voltage side is defined as 30 degrees ahead of the low-voltage side for the positive-sequence network. Therefore, the D1-Yg transformer connection is selected for the final model. This selected D1-Yg connection is then used in the following short-circuit calculations in Task 3 and Task 4.

5. Task 3 – Three-Phase Fault Analysis

In this section, a three-phase short-circuit fault at BUS3 is analysed. The calculation is based on the pre-fault load flow results obtained in Section 4. Since a three-phase fault is a balanced fault, only the positive-sequence network is required. The fault current is calculated using the Thevenin equivalent circuit seen from BUS3, and the bus voltages and branch currents during the fault are obtained using the impedance matrix method and the superposition principle.

5.1 Equivalent Load Admittance

Before carrying out the short-circuit calculation, the constant-power loads are converted into equivalent constant admittances. This is because the network admittance matrix requires each load to be represented as an admittance connected to the corresponding bus.

For each PQ load bus, the complex load power is:

$$S_{load} = P_{load} + jQ_{load}$$

$$S_{load} = 0.5 + j0.3\ pu$$

The equivalent load admittance is calculated from the complex power and the pre-fault bus voltage magnitude:

$$Y_{load,i} = \frac{S_{load}^*}{|V_{pre,i}|^2}$$

where $|V_{pre,i}|$ is the pre-fault voltage magnitude at bus i obtained from the load flow result. The complex conjugate of the load power is used because the complex power relationship is:

$$S = VI^*$$

Therefore:

$$Y = \frac{I}{V} = \frac{S^*}{|V|^2}$$

The calculated equivalent load admittances are summarised in Table 8.

Table 8. Equivalent load admittances used in the three-phase fault calculation.

Bus	Pre-fault voltage magnitude	Equivalent load admittance
BUS2	0.9246 pu	0.5849 − j0.3509 pu
BUS3	0.9183 pu	0.5929 − j0.3558 pu
BUS4	0.9167 pu	0.5929 − j0.3558 pu
BUS5	0.9196 pu	0.5950 − j0.3570 pu
BUS6	0.9259 pu	0.5832 − j0.3499 pu

5.2 Thevenin Equivalent at BUS3

A three-phase short-circuit fault is applied at BUS3.To calculate the short-circuit current, the Thevenin equivalent circuit seen from BUS3 is developed using the bus impedance matrix method.

The Thevenin open-circuit voltage is taken as the pre-fault voltage at BUS3 from the load flow result:

$$V_{th} = V_{pre,3}$$

$$V_{th} = 0.9183\angle 25.4490° \; pu$$

The Thevenin equivalent impedance is obtained from the diagonal element of the positive-sequence bus impedance matrix corresponding to BUS3:

$$Z_{th} = Z_{33}$$

$$Z_{th} = 0.017531 + j0.082250 \; pu$$

The three-phase fault current is then calculated by:

$$I_{fault} = \frac{V_{th}}{Z_{th}}$$

Substituting the calculated values gives:

$$I_{fault} = \frac{0.9183\angle 25.4490°}{0.017531 + j0.082250}$$

$$I_{fault} = 10.9194\angle(-52.5190°) \; pu$$

The current base at the 230 kV voltage level is:

$$I_{base,230kV} = \frac{S_{base}}{\sqrt{3}V_{base,230kV}}$$

$$I_{base,230kV} = \frac{100 \; MVA}{\sqrt{3} \times 230 \; kV}$$

$$I_{base,230kV} = 251.0219 \; A$$

Therefore, the actual fault current is:

$$I_{fault,actual} = |I_{fault}|I_{base,230kV}$$

$$I_{fault,actual} = 10.9194 \times 251.0219$$

$$I_{fault,actual} = 2741.02\ A$$

$$I_{fault,actual} = 2.7410\ kA$$

```
Task 3 three-phase fault at BUS_3:
Vth = 0.9183 angle 25.4490 deg pu
Zth = 0.017531 +0.082250j pu
Ifault = 10.9194 angle -52.5190 deg pu
Ibase at 230 kV = 251.02 A
Fault current magnitude = 2741.02 A = 2.7410 kA
```

Figure 21. MATLAB calculation output for the BUS3 three-phase fault current.

5.3 Bus Voltages and Branch Currents During Fault

The bus voltages during the three-phase fault are calculated using the superposition principle. The faulted network voltage change is obtained from the bus impedance matrix and the fault current. The voltage at each bus during the fault is calculated as:

$$V_i = V_{pre,i} - Z_{i3} I_{fault}$$

where Z_{i3} is the element in row i and column 3 of the bus impedance matrix. Since the fault is applied directly at BUS3, the voltage magnitude at BUS3 becomes zero. Buses electrically close to BUS3 experience larger voltage drops, while buses further away from the fault point have higher remaining voltage magnitudes.

The calculated bus voltages during the three-phase fault are summarised in Table 9.

Table 9. Bus voltages during the three-phase fault at BUS3.

Bus	Voltage magnitude	Voltage angle
BUS1	0.6124 pu	-19.2752 deg
BUS2	0.0668 pu	25.5072 deg
BUS3	0.0000 pu	0.0000 deg
BUS4	0.1500 pu	30.5388 deg
BUS5	0.2648 pu	30.6902 deg
BUS6	0.3093 pu	31.0971 deg
BUS7	0.6673 pu	-7.8780 deg

The branch currents are calculated from the voltage difference across each branch impedance. For a branch connected between bus i and bus j, the branch current is calculated as:

$$I_{ij} = \frac{V_i - V_j}{Z_{ij}}$$

The calculated branch currents during the BUS3 three-phase fault are summarised in Table 10.

Table 10. Branch currents during the three-phase fault at BUS3.

Branch	Current magnitude	Current angle
T1 BUS1–BUS2	5.6690 pu	-114.0384 deg
Line1 BUS2–BUS3	4.6544 pu	-55.4025 deg
Line2 BUS3–BUS4	6.2687 pu	129.6291 deg
Line3 BUS4–BUS5	2.9991 pu	129.9782 deg
Line4 BUS5–BUS6	3.0991 pu	132.6102 deg
Line5 BUS4–BUS6	3.3283 pu	130.7132 deg
T2 BUS6–BUS7	9.3809 pu	57.6183 deg

5.4 Simulink Fault Simulation

To verify the analytical calculation, a Three-Phase Fault block was connected at BUS3 in the Simulink model. The fault block was configured as a three-phase-to-ground fault by selecting Phase A, Phase B, Phase C and Ground. The fault switching time was set to 1 s, so the pre-fault and post-fault responses could be observed in the same scope window.

The fault block settings are shown in Figure 22.

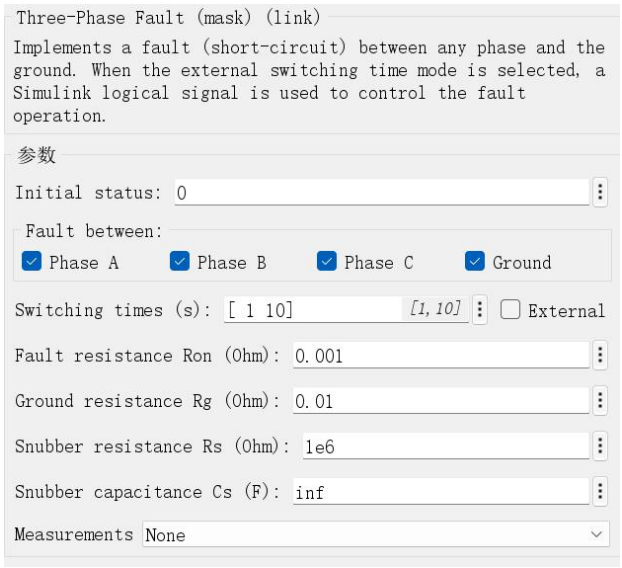


Figure 22. Three-phase fault block settings for the BUS3 fault.

The simulated BUS3 voltage magnitude is shown in Figure 23. Before the fault, the BUS3 voltage is approximately 0.9183 pu. After the fault is applied at 1 s, the voltage magnitude drops to approximately zero, which agrees with the theoretical calculation.

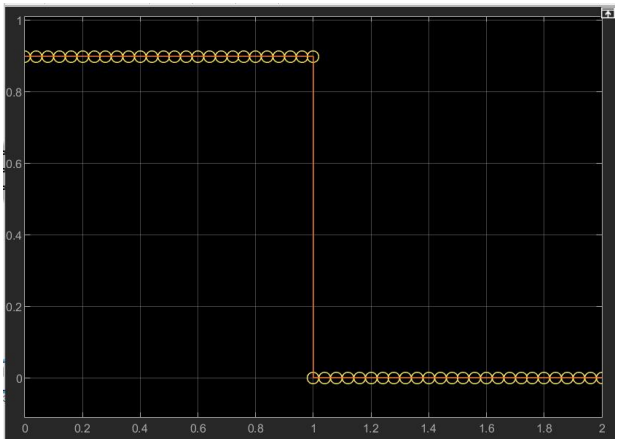


Figure 23. Simulated BUS3 voltage magnitude during the three-phase fault.

The simulated fault current magnitude is shown in Figure 24. After the fault is applied, the current increases sharply to approximately 10.9 pu. This is close to the calculated fault current magnitude of 10.9194 pu.

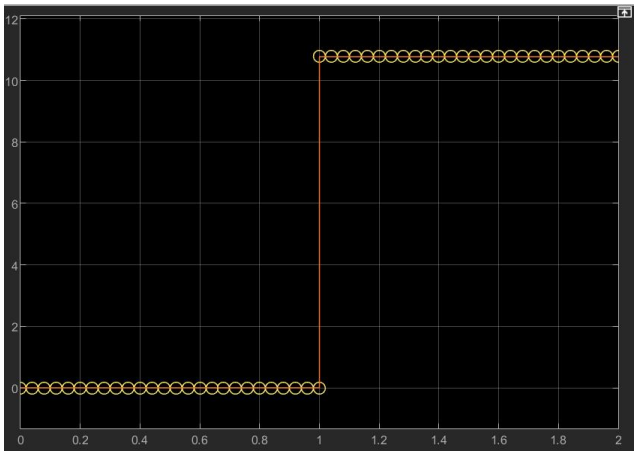


Figure 24. Simulated BUS3 fault current magnitude during the three-phase fault.

The analytical and simulation results are compared in Table 11.

Table 11. Comparison between analytical and simulation results for the BUS3 three-phase fault.

Quantity	Analytical result	Simulink result	Comment
BUS3 pre-fault voltage	0.9183 pu	approximately 0.918 pu	Good agreement
BUS3 voltage after fault	0.0000 pu	approximately 0 pu	Good agreement
Fault current magnitude	10.9194 pu	approximately 10.9 pu	Good agreement
Fault current actual value	2.7410 kA	close to 2.74 kA	Engineering agreement

The comparison shows that the Simulink simulation agrees well with the analytical calculation. The voltage at BUS3 collapses to zero after the fault, and the simulated fault current magnitude is close to the calculated value. Small differences may be caused by

numerical rounding, phasor-domain modelling and the non-zero fault resistance used in the Three-Phase Fault block. Therefore, the analytical calculation and simulation output are considered consistent in an engineering sense.

6. Task 4 – Phase-A-to-Ground Fault Analysis

In this section, a phase-A-to-ground fault at BUS3 is analysed. Unlike the balanced three-phase fault in Section 5, a single-phase-to-ground fault is an unbalanced fault. Therefore, the positive-, negative- and zero-sequence networks are required. The sequence Thevenin equivalent impedances are first obtained from the corresponding bus impedance matrices. Then, the sequence currents, phase currents and phase voltages are calculated and compared with the Simulink simulation results.

6.1 Sequence Thevenin Equivalent Impedances

For a phase-A-to-ground fault at BUS3, the positive-, negative- and zero-sequence Thevenin equivalent impedances seen from BUS3 are required. These impedances were obtained using the nodal impedance matrix method.

The current base at the 230 kV voltage level is:

$$I_{base,230kV} = \frac{S_{base}}{\sqrt{3}V_{base,230kV}}$$

$$I_{base,230kV} = \frac{100\ MVA}{\sqrt{3} \times 230\ kV}$$

$$I_{base,230kV} = 251.0219\ A$$

The Thevenin open-circuit voltage is taken from the pre-fault load flow result at BUS3:

$$V_{th} = V_{pre,3}$$

$$V_{th} = 0.9183\angle 25.4490^{\circ}\ pu$$

The calculated sequence Thevenin impedances are summarised in Table 12.

Table 12. Sequence Thevenin equivalent impedances seen from BUS3.

Sequence network	Thevenin impedance
Positive sequence	0.017531 + j0.082250 pu
Negative sequence	0.019662 + j0.086908 pu
Zero sequence	0.012207 + j0.076835 pu

The positive-sequence Thevenin impedance is the same as the Thevenin impedance used in the three-phase fault calculation. The negative- and zero-sequence impedances are different

because the generator, transformer and line sequence parameters are different in the negative- and zero-sequence networks.

6.2 Sequence Currents and Phase Fault Currents

For a phase-A-to-ground fault, the positive-, negative- and zero-sequence networks are connected in series. Therefore, the three sequence currents are equal:

$$I_1 = I_2 = I_0$$

The sequence current is calculated as:

$$I_1 = I_2 = I_0 = \frac{V_{th}}{Z_{1,th} + Z_{2,th} + Z_{0,th}}$$

Substituting the calculated sequence impedances gives:

$$I_1 = I_2 = I_0 = 3.6600\angle(-53.1958°)\ pu$$

The phase currents are obtained from the sequence currents using symmetrical component transformation. For a phase-A-to-ground fault:

$$I_a = I_1 + I_2 + I_0$$

$$I_b \approx 0$$

$$I_c \approx 0$$

Therefore:

$$I_a = 10.9799\angle(-53.1958°)\ pu$$

$$I_b \approx 0\ pu$$

$$I_c \approx 0\ pu$$

The actual phase-A fault current is:

$$I_{a,actual} = |I_a| I_{base,230kV}$$

$$I_{a,actual} = 10.9799 \times 251.0219$$

$$I_{a,actual} = 2756.19\ A$$

$$I_{a,actual} = 2.7562\ kA$$

Table 13. Sequence and phase currents for the BUS3 phase-A-to-ground fault.

Quantity	Magnitude	Angle
I1	3.6600 pu	-53.1958 deg
I2	3.6600 pu	-53.1958 deg
I0	3.6600 pu	-53.1958 deg
Ia	10.9799 pu	-53.1958 deg
Ib	approximately 0 pu	-
Ic	approximately 0 pu	-
Ia actual value	2.7562 kA	-

6.3 Phase Voltages During the A-G Fault

The phase voltages during the phase-A-to-ground fault were calculated using the superposition principle and the bus impedance matrix. At the faulted bus, the phase-A voltage becomes zero because phase A is directly connected to ground through the fault. The unfaulted phase voltages remain non-zero because phases B and C are not directly connected to ground.

The phase voltages at BUS3 and BUS1 are summarised in Table 14.

Table 14. Phase voltages at BUS3 and BUS1 during the phase-A-to-ground fault.

Bus	Va magnitude	Va angle	Vb magnitude	Vb angle	Vc magnitude	Vc angle
BUS3	0.0000 pu	90.0000 deg	0.9336 pu	-92.0090 deg	0.8994 pu	143.5058 deg
BUS1	0.7086 pu	-11.3264 deg	0.8830 pu	-112.7404 deg	1.0170 pu	110.3399 deg

The calculated result shows that $|V_{a,3}|$ is zero during the fault, which is the expected result for a phase-A-to-ground fault at BUS3. At BUS1, the phase voltages are not zero because BUS1 is not directly faulted, but the unbalanced fault still affects the voltage magnitudes and phase angles through the sequence networks.

6.4 Simulink Simulation Verification

To verify the analytical calculation, the Three-Phase Fault block was configured as a phase-A-to-ground fault. Only Phase A and Ground were selected in the fault block. The switching time was set to 1 s, so the pre-fault and post-fault responses could be observed clearly.

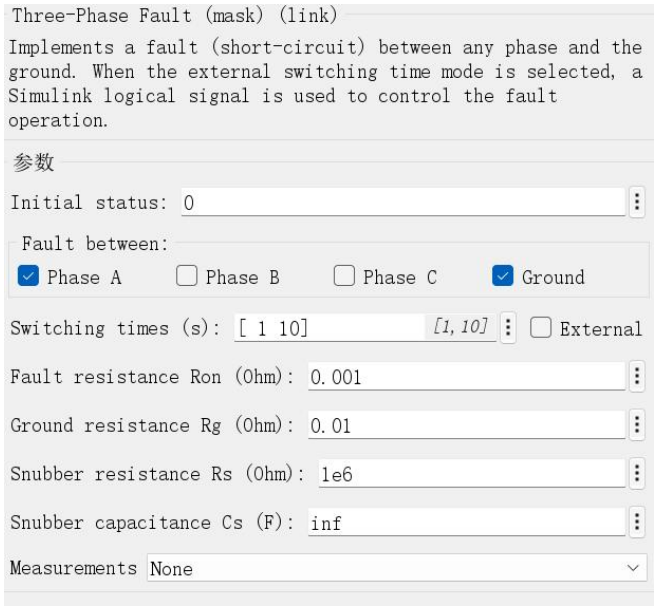


Figure 25. Phase-A-to-ground fault block settings at BUS3.

The simulated BUS3 phase voltage magnitudes are shown in Figure 26. Before the fault, the three phase voltage magnitudes are close to the pre-fault operating voltage. After the fault is applied at 1 s, the phase-A voltage drops to approximately zero, while the phase-B and phase-C voltages remain non-zero. This agrees with the analytical result in Table 14.

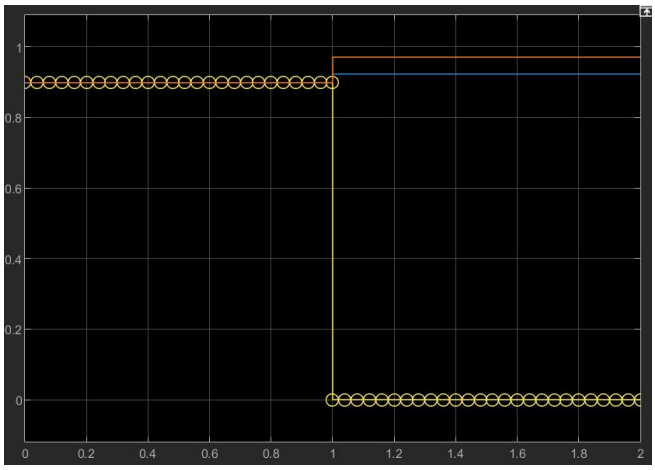


Figure 26. Simulated BUS3 phase voltage magnitudes during the phase-A-to-ground fault.

The simulated fault current magnitudes are shown in Figure 27. The result shows that the main fault current appears in phase A, while the phase-B and phase-C currents are close to zero. This is consistent with the theoretical condition of a phase-A-to-ground fault.

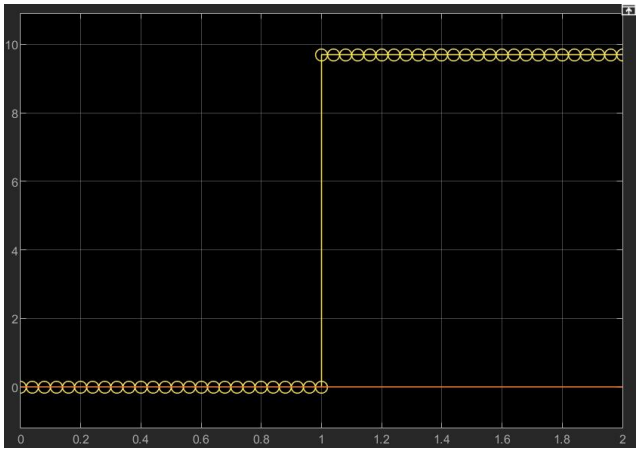


Figure 27. Simulated phase fault current magnitudes during the phase-A-to-ground fault.

The analytical and simulation results are compared in Table 15.

Table 15. Comparison between analytical and simulation results for the BUS3 phase-A-to-ground fault.

Quantity	Analytical result	Simulink result	Comment
BUS3 phase-A voltage after fault	0.0000 pu	approximately 0 pu	Good agreement
BUS3 phase-B voltage after fault	0.9336 pu	approximately 0.93 pu	Good agreement
BUS3 phase-C voltage after fault	0.8994 pu	approximately 0.90 pu	Good agreement
Phase-A fault current magnitude	10.9799 pu	approximately 11 pu	Good agreement
Phase-A fault current actual value	2.7562 kA	close to 2.76 kA	Engineering agreement
Phase-B and phase-C fault currents	approximately 0 pu	close to 0 pu	Good agreement

7. Discussion

This lab combines load flow analysis, symmetrical fault analysis and unsymmetrical fault analysis in one seven-bus power system model. The key purpose is not only to obtain numerical results, but also to verify whether the analytical calculation and the Simulink simulation describe the same physical behaviour.

From the load flow study, the D1-Yg, D11-Yg and Yg-Yg transformer connection cases produce almost the same bus voltage magnitudes and generator power outputs. However, the bus voltage phase angles change significantly. Compared with the Yg-Yg case, the D1-Yg connection shifts the voltage angles of the transmission-side buses by approximately +30 degrees, while the D11-Yg connection shifts them by approximately -30 degrees. This confirms that the transformer vector group mainly affects the voltage phase-angle relationship rather than the steady-state voltage magnitude profile.

For the three-phase fault at BUS3, the calculated fault current is:

$$I_{fault} = 10.9194\angle(-52.5190°)\ pu$$

The corresponding actual current at the 230 kV level is:

$$I_{fault,actual} = 2.7410\ kA$$

The calculated BUS3 voltage becomes zero during the fault, which is physically expected because BUS3 is directly short-circuited. The Simulink voltage scope also shows that the BUS3 voltage drops from the pre-fault value of about 0.9183 pu to approximately zero after the fault is applied at 1 s. The simulated fault current magnitude is also close to the analytical value. Therefore, the analytical calculation and simulation result agree well for the three-phase fault.

For the phase-A-to-ground fault at BUS3, the positive-, negative- and zero-sequence networks are all required. The calculated sequence currents are equal:

$$I_1 = I_2 = I_0 = 3.6600\angle(-53.1958°)\ pu$$

The corresponding phase currents are:

$$I_a = 10.9799\angle(-53.1958°)\ pu$$

$$I_b \approx 0\ pu$$

$$I_c \approx 0\ pu$$

The actual phase-A fault current is:

$$I_{a,actual} = 2.7562\ kA$$

This result is consistent with the theoretical behaviour of a phase-A-to-ground fault. Since only phase A is directly connected to ground, the phase-A current is large, while the phase-B and phase-C fault currents are approximately zero. The calculated phase voltage at BUS3 also confirms this behaviour:

$$|V_{a,3}| = 0.0000\ pu$$

$$|V_{b,3}| = 0.9336\ pu$$

$$|V_{c,3}| = 0.8994\ pu$$

The Simulink simulation shows the same trend: the phase-A voltage drops to approximately zero, while the other two phase voltages remain non-zero. Therefore, the analytical sequence-network calculation is consistent with the simulation output.

Small differences between the analytical and simulation results are acceptable. The main reasons are numerical rounding, the non-zero fault resistance and ground resistance used in the Three-Phase Fault block, the phasor-domain modelling assumptions, and the conversion of constant-power loads into equivalent admittances. In engineering practice, these differences are small enough that the analytical and simulation results can be considered consistent.

Overall, the results show that the per-unit system, load flow calculation, bus impedance matrix method and sequence-network method are correctly applied. The simulation results support the theoretical calculations for both the balanced three-phase fault and the unbalanced phase-A-to-ground fault.

8. Conclusion

In this lab, a seven-bus power system was modelled in Simulink using the Specialized Power Systems toolbox. A common three-phase base power of 100 MVA was used, and component parameters of generators, transformers, transmission lines and loads were converted into consistent per-unit values. The completed model was then used for load flow analysis, three-phase fault analysis and phase-A-to-ground fault analysis.

The load flow study provided the pre-fault operating point of the system. The final D1-Yg case gave a BUS3 pre-fault voltage of:

$$V_{pre,3} = 0.9183\angle 25.4490^\circ \; pu$$

This value was used as the Thevenin open-circuit voltage for the BUS3 fault calculations. The comparison of transformer winding connections showed that D1-Yg, D11-Yg and Yg-Yg connections gave almost the same voltage magnitudes, but different voltage phase angles. The D1-Yg connection was selected for the final model because it follows the North American standard used in the course.

For the three-phase fault at BUS3, the Thevenin equivalent impedance was:

$$Z_{th} = 0.017531 + j0.082250 \; pu$$

The calculated three-phase fault current was:

$$I_{fault} = 10.9194\angle(-52.5190^\circ) \; pu$$

This corresponds to an actual fault current of:

$$I_{fault,actual} = 2.7410 \; kA$$

The bus voltage calculation and Simulink simulation both showed that the BUS3 voltage drops to approximately zero during the three-phase fault.

For the phase-A-to-ground fault at BUS3, the calculated sequence Thevenin impedances were:

$$Z_{1,th} = 0.017531 + j0.082250 \; pu$$

$$Z_{2,th} = 0.019662 + j0.086908 \; pu$$

$$Z_{0,th} = 0.012207 + j0.076835 \; pu$$

The calculated phase-A fault current was:

$$I_a = 10.9799\angle(-53.1958^\circ) \; pu$$

This corresponds to:

$$I_{a,actual} = 2.7562 \; kA$$

The calculated and simulated phase voltages confirmed that the phase-A voltage at BUS3 becomes zero, while the phase-B and phase-C voltages remain non-zero.

Overall, the analytical results and Simulink simulation results are consistent in an engineering sense. This confirms that the system modelling, per-unit conversion, load flow calculation, Thevenin equivalent method, bus impedance matrix method and sequence-network method were correctly applied in this lab.